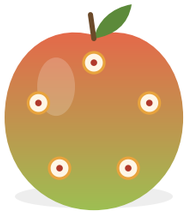
REFERENCE-VALUE UNCERTAINTY IN NIR FRUIT CALIBRATION

soluble solids & dry matter · replicate-sampling design rule · attenuation diagnostic · leakage-audited validation

apples 633 × 5 faces · kiwifruit 4,318 × 2–5 devices

1 REFERENCE SIDE



- apples × flesh faces **633 × 5**
- median 5-face range **2.80 °Brix**
- within-fruit SD vs meter **5.6 × spec (±0.2)**

one fruit, five answers

2 ERROR BUDGET

$Y_{ij} = \mu + a_i + e_{ij}$ $\quad \sigma a^2 = 2.327 \quad \sigma f^2 = 2.551$

reference variance **ICC 0.477 [0.436, 0.515]**

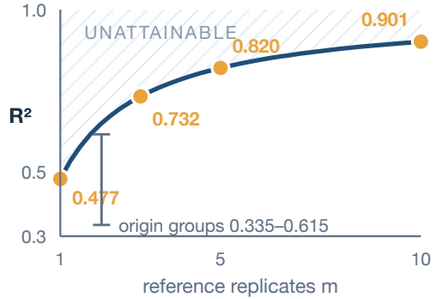
52.3% within	47.7% between

$RMSEP^2 = model + \sigma f^2/m + instrument$

model	reference	instrument

two of the three parts are not the model's — decompose before tuning models

3 ATTAINABLE CEILING



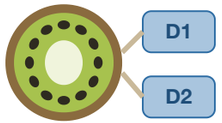
4 DESIGN RULE

$m \geq (\sigma f^2/\sigma a^2) \cdot R^2/(1-R^2)$

target R^2	faces m ≥
0.70	3
0.80	**5**
0.90	10
0.95	21

$\sigma f^2/\sigma a^2 = 1.0965$ from a pilot batch · *lower bound — leave a margin*

5 INSTRUMENT SIDE (KIWIFRUIT)



SNV-corrected spectral variance

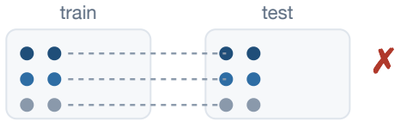
52.6%	17.1%	30.3%

between fruit · device · fruit×device + scan noise

pooled fruit×device/scan error ≈ 0.505 %DM — spans 20.6% of its mean over ten points

yet its share of MSE rises 4.3% → 24.7% (×5.7) as R^2 increases

6 VALIDATION AUDIT — SPLIT LEAKAGE



✗

R^2 0.1692 random split — inflated

R^2 0.1195 grouped — honest

face-level random splits inflate R^2 1.41–1.46×

7 FALSIFIABLE ATTENUATION DIAGNOSTIC

six parameter-free predictions, all close to measurement — two shown

	pred	meas	
label-only baseline R^2	0.3462	**0.3456**	✓
attenuation ratio	0.5816	**0.5636**	✓

REPORTED R^2 SHOULD BE READ AGAINST A REFERENCE-LIMITED CEILING, NOT AGAINST 1 — A PILOT SUBSET GIVES THAT CEILING BEFORE THE EXPERIMENT